Maxime Newman Nereyabagabo

Burnaby, BC | (778) 814-0670 | maximenewman05@gmail.com | LinkedIn | GitHub | Website

Education

Simon Fraser University Jan. 2024 – Aug. 2027
Bachelor of Science, Computing Science (Minor in Statistics) *Burnaby, BC*

- Coursework: Machine Learning, Data Structures, Algorithms, Databases, Statistics, Networking, Systems Programming

Technical Skills

Languages & Scripting: Python, Bash, Java, JavaScript, C, C++, C#, HTML/CSS, SQL
Cloud, DevOps & Automation: Docker, GitHub Actions, CI/CD, AWS, GCP, Azure, Fly.io, Git, Linux/Unix
Databases & Caching: PostgreSQL, MySQL, Redis, Supabase
ML & AI: Scikit-learn, LangGraph, Pydantic, Pandas, NumPy, LangSmith
Frameworks & Libraries: Next.js, FastAPI, Django
Testing & Monitoring: Pytest, JUnit, Mockito, GoogleTest, LangSmith eval/regression monitoring

Relevant Experience

SFU Cybersecurity | GitHub May. 2026 – Present
Research Assistant *Burnaby, BC*

- Contributing to an agentic security pipeline that autonomously detects, exploits, patches, and verifies vulnerabilities using **LLMs**
- Primary owner of end-to-end pipeline runs across 120 **Java** projects from **CWE-Bench-Java**, producing the scans, exploits, and patches behind the paper's evaluation.
- Improved pipeline reliability to run unattended across the full 120-project benchmark by adding graceful fallback that reuses cached scan results and runs fresh analysis otherwise, eliminating prior startup failures.
- Documenting this work as a formal research manuscript intended for submission to the ACM Conference on Computer and Communications Security (CCS), emphasizing system reliability, verification, and auditability

SKC Engineering Ltd Jan. 2026 – May 2026
Software Developer (Dev Ops/Data Engineer) *Surrey, BC*

- Migrated **29,000+ rows** across **87 tables** from legacy Linode MySQL to managed DigitalOcean MySQL with **100% parity** by engineering a topological-sort foreign-key resolver, idempotent batched inserts, and symmetric pre/post-migration primary-key validation
- Saved an estimated **$15,000+** in engineer time by building structured CSV logging with company-level and file-level outcomes, enabling full auditability and data quality tracking across parallel migration runs
- Built an authenticated PowerShell pipeline via Azure App Registration to classify **15,639 SharePoint files** into **6 categories** with **70% requiring no manual review**, delivering a cleaned dataset to the ML team

SKC Engineering Ltd Sept. 2025 – Jan. 2026
Software Developer (AI/ML Engineer) *Surrey, BC*

- Achieved **sub-300 ms API response times** for core authenticated endpoints by deploying to Fly.io using **Docker** multi-stage builds, **GitHub Actions CI/CD**, JWT-based access control, and idempotent APIs supporting a persistent multi-user workspace
- Built a full-stack AI-powered cost estimation platform with **GPT-4o mini** and **LangGraph**, using intent-based routing, deterministic tool execution, and structured state management to deliver consistent, auditable outputs
- Improved model consistency across deployments with **50+ LangSmith eval cases**, using human-in-the-loop interrupts and a regression monitoring pipeline to detect behavioral drift
- Cut LLM inference cost per request by **20%** by stabilizing input size across long sessions through rolling context windows, automatic summarization, and state-first prompting

SFU CS ARCH Group Jan 2025 – Present
Research Assistant *Burnaby, BC*

- Designed GPU-accelerated matrix multiplication kernels in **CUDA** on Linux, achieving **93.7% of cuBLAS** performance, using NVIDIA Nsight Compute (NCU) for profiling and inspecting PTX assembly to validate compiler IR passes
- Researched attention mechanisms and transformer architectures, focusing on how large-scale matrix multiplications dominate training and inference of LLMs
- Authored a 'read-along guide' on GPU multi-threading and execution models (thread blocks, warps, SM scheduling) for kernel optimization

Experience Continued

Zebra Robotics Surrey Jan. 2025 – Jun. 2025
Robotics Instructor *Surrey, BC*

- Taught programming and robotics fundamentals using Python, Arduino, and microcontroller labs, enabling students to program motors and sensors and complete functional robot prototypes
- Led student robotics projects from concept through testing, guiding teams in mechanical iteration and hardware debugging, resulting in completed robots showcased in a school exhibition

Projects

Goblin's Keep | *Java, Maven, JUnit, Mockito, GitHub Actions* GitHub | Play Game

- Led backend development of a 2D tile-based escape game, architecting game logic and AI systems, establishing **CI/CD pipelines with GitHub Actions**, and coordinating code reviews across a team of developers
- Modified **A* pathfinding algorithm** with randomized patrol logic, enhancing difficulty balance and replayability
- Achieved **96% line coverage** and **92% branch coverage** with a **JUnit/Mockito** automated test suite ensuring correctness across all gameplay interactions

FraudLens | *Python, Pandas, Scikit-learn, NumPy* GitHub | SFU DSSS ML Hackathon

- **Won 1st place** at the SFU DSSS ML Hackathon by building a multi-class fraud detection pipeline using Python, Pandas, and scikit-learn under tight time constraints
- **Delivered a 228% macro F1 improvement** over the logistic regression baseline (0.994 vs 0.303) by training a class-weighted Random Forest on imbalanced data under tight inference latency constraints
- Engineered 5+ predictive features using Pandas and NumPy, including **balance mismatch flags and error signals** to expose accounting inconsistencies, improving recall on rare minority fraud classes
- Built a constraint-based data validation pipeline that enforced consistent transaction types and flagged impossible balances, improving model robustness and interpretability for fraud analysis

Other Experiences

Simon Fraser University Jan. 2025 – Apr. 2025
Calculus Teaching Assistant *Burnaby, BC*

- Led weekly peer-led sessions for a dedicated group of students, fostering collaborative learning and long-term academic growth
- Explained key concepts through interactive discussions and guided problem-solving sessions
- Reinforced personal mastery of calculus through teaching, feedback, and real-time troubleshooting

Simon Fraser University Aug. 2024 – Sept. 2024
Hive Leader *Burnaby, BC*

- Organized and led events to welcome and engage new students
- Collaborated with other Hive Leaders to plan and execute group activities
- Supported and mentored HIVE volunteers aspiring to become Hive Leaders
- Led orientation sessions, ensuring new students had a smooth transition

SFU Recreation Apr. 2024 – Aug. 2024
Recreation Assistant *Burnaby, BC*

- Front Desk Reception: Managed check-ins, answered inquiries, and kept daily operations running smoothly
- Customer Service: Built relationships with clients, addressed concerns, and made sure every member felt welcomed
- Fitness Class Assistant: Supported instructors with class setup and equipment while keeping participants motivated and sessions running safely
- Intramural Soccer League Assistant: Helped organize and run a soccer league, handling scheduling, officiating, and ensuring games ran smoothly